

Fungal Skin Infections

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Fungal Skin Infections

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Introduction

Spores of fungi occur in the **hot and humid or arid tropical environment**.

Poor personal and community hygiene, closeness among large families, and institutions such as schools, camps and barracks favour the spread of fungal infections among children.

Most of these fungal infections affect the **skin, hair, nails and mucosa**.

Classification of fungal skin infections

Superficial

- Candidiasis
- Tinea capitis
- Tinea corporis
- Tinea pedis
- Pityriasis versicolor

Deep

- Mycetoma (Madura foot)
- Blastomycosis
- Coccidiomycosis
- Histoplasmosis
- Sporotrichosis

Candidiasis

An infection caused by *Candida albicans*, *C. tropicalis*, *C. stellatoidea*, *C. krusei*, or other yeasts of the genus *Candida*.

The infections usually occur on the **skin, mucous membranes, nail folds and nails, gastrointestinal tract and genital tract**. In systemic candida infections, the **viscera** may be involved.

The **skin infection** may present as **dusky red macerated or erosive lesions** in the skin folds of the **neck, flanks, large flexures and limbs**. Fissuring and bleeding may follow the scratching of affected areas, with occurrence of **secondary bacterial infection**.

Oral candidiasis (thrush)

Caused by yeast-like fungi of the genus *Candida* – most commonly *Candida albicans*.

Characterised by **curdy or whitish-grey, friable, cheesy pseudomembranous patches or plaques on markedly reddened mucosa**.

The plaques of thrush **invade the mucosa superficially** and may be found on the **lips, buccal mucosa, tongue, and palate**, extending to the **oesophagus/pharynx**.

Removal of plaques from these surfaces may cause mild punctate areas of bleeding, which helps to confirm the diagnosis.

Predisposing factors to candidiasis:

1. Diabetes mellitus
2. Retroviral disease
3. Prolonged antibiotic and corticosteroid use
4. Leukaemia, etc.

Treatment:

- Antifungal drugs such as **clotrimazole, nystatin, amphotericin B, miconazole gel**
- Proper oral hygiene

Tinea capitis

A **dermatophyte infection of the scalp**, most often caused by *Trichophyton tonsurans*, occasionally by *Microsporum canis*, and much less commonly by other *Microsporum* and *Trichophyton* species.

Lesions occur as **erythematous and scaly circular plaques (ringworm)**, within which the infected hairs become **brittle and broken with different lengths**.

Multiple affected sites produce a **moth-eaten appearance** of hair.

Scaling and inflammation of the scalp are variable. Numerous **confluent patches of alopecia** develop, and patients may complain of **severe pruritus**.

Kerion

Sometimes the host's **cellular immune response** to tinea capitis may lead to the development of an **inflammatory and infected boggy mass covered in crusts, pus-filled bumps and localized hair loss**, referred to as a **kerion**.

Favus

A **severe, chronic form** of tinea capitis, typically caused by the dermatophyte *Trichophyton schoenleinii*.

Characterised by **thick, yellowish, cup-shaped yellow crusts called scutula** that develop around hair follicles, often accompanied by a **distinctive mousy odour**.

Also termed **honey-combed scalp lesions** with yellow crusts and scales.

If left untreated, it **destroys the hair follicles**, causes **matting of the hair** and **permanent scarring hair loss (alopecia)**.

Diagnosis and treatment

Diagnosis: Scraping and **KOH examination**.

Treatment:

- Oral **griseofulvin, itraconazole or terbinafine**
- Topical antifungal cream, shampoo

Tinea corporis

A fungal infection of the **glabrous skin**, excluding the palms, soles and groin.

Most common aetiologic organisms:

- *Trichophyton rubrum*
- *Trichophyton mentagrophytes*
- Infections with *Microsporum canis* are also frequent

Characteristic appearance: a **dry, mildly erythematous, elevated, scaly papule or plaque** that **spreads centrifugally and clears centrally** to form the characteristic **annular lesion**. The **active edge shows vesicles and pustules**.

It is usually acquired by **direct contact from an infected person, surfaces, and pets (*M. canis*)**.

Other forms by site

1. **Tinea faciale** – infection of the glabrous skin localized to the face
2. **Tinea axillaris** – infection of the axilla
3. **Tinea cruris** – infection of the groin
4. **Tinea manuum** – infection of the hands
5. **Tinea unguium** – infection of the nails

Diagnosis and treatment

Diagnosis: clinical; skin scraping and **KOH examination**; **culture of skin scrapings** for identifying the species of dermatophyte.

Treatment:

1. **Topical antifungal agents** – e.g. imidazoles, terbinafine, naftifine
2. In severe cases, **oral griseofulvin, itraconazole**

Tinea pedis (athlete's foot)

A fungal infection of the **toe webs and soles of the feet**.

Usual etiologic agents:

1. *Trichophyton rubrum*
2. *Trichophyton mentagrophytes*
3. *Epidermophyton floccosum*

Features:

- The **subdigital crevices are fissured**, with **maceration and peeling** of the surrounding skin
- Less commonly, a **chronic diffuse hyperkeratosis of the sole** occurs, with only mild erythema
- Occasionally, the papules progress to **vesicles and bullae** that may become pustular
- **Severe tenderness, itching, and a persistent foul odour** are characteristic

Diagnosis and treatment

Diagnosis: fungal mycelia seen on microscopic examination of a **KOH preparation**, or by culture.

Treatment:

1. **Avoidance of occlusive footwear**; careful **drying between the toes** after bathing
2. Topical **zinc undecylenate powder**, **imidazole cream**, oral **griseofulvin**

Pityriasis versicolor

A common **chronic fungal infection of the stratum corneum**.

Caused by the **dimorphic yeast** *Malassezia furfur*, found in areas of skin that are **rich in sebum production**.

The disease is most prevalent in **adolescents and young adults**.

Predisposing factors:

- A warm humid environment
- Excessive sweating

- Occlusion
- High plasma cortisol levels
- Immunosuppression
- Malnourishment
- Genetically determined susceptibility

Lesions: either **hypopigmented or hyperpigmented macules** covered with a fine scale, which often **merge to form confluent patches**.

Sites: most commonly the **neck, upper chest, back, and upper arms**.

Diagnosis: **Wood lamp**, KOH preparation of scrapings, skin biopsy.

Treatment:

- **Selenium sulfide suspension** applied overnight
- **Imidazole or terbinafine cream** twice daily for 2-4 weeks
- Oral therapy with **ketoconazole, fluconazole, or itraconazole**

Histoplasmosis

A **chronic infection** caused by *Histoplasma capsulatum* and *H. duboisii*, dimorphic fungi acquired by **inhalation of fungal spores**.

These are found in the **soil and in the droppings of some animals** e.g. bats, poultry.

The **primary infection is in the lungs**, with **haematogenous spread** to the **liver, bone marrow, spleen, lymph nodes and subcutaneous tissue**.

Skin lesions appear as **papules, pustules and nodules**, which may be found throughout the body and are caused by the fungus spreading to infect the skin.

H. duboisii (African histoplasmosis) presents as **subcutaneous nodules which tend to umbilicate and suppurate or ulcerate** as they grow.

H. capsulatum characteristically affects the **lungs**, and may present with **productive cough and constitutional symptoms**.

Diagnosis: isolation and identification of the causative fungus in cultures from **sputum, bone marrow, lymph node biopsy, or pus or scrapings from ulcers**.

Treatment: IV amphotericin B; oral itraconazole or ketoconazole.

Sporotrichosis

A chronic infection which may follow **accidental implantation into the skin** of *Sporothrix schenckii*, which **grows on thorny plants**.

The infection typically begins as a **firm, painless or mildly tender subcutaneous nodule (bump) at the site of skin trauma**, and presents as a **purple, violaceous, hyperpigmented brown, or dark grey nodule**.

The primary nodule **grows larger, becomes pustular, and may open into a chronic, hollowed-out sore (ulceration)**.

Early lesions are **indurated subcutaneous nodules at the sites of implantation**.

Infection may spread by **lymphatic or haematogenous routes** to muscles, bones, joints or other tissues. **Inhalation of spores may cause lung infection.**

Diagnosis: microscopic demonstration of the fungus in preparations from scrapings or biopsy.

Treatment: potassium iodide, IV amphotericin B, oral itraconazole.

Blastomycosis

A chronic infection caused by *Blastomyces dermatitidis*, acquired by **inhalation of fungal spores.**

The **primary infection is in the lungs**, with **haematogenous spread to the skin, bone and genital tract.**

Presents as **irregularly crusted plaques, superficial ulcerative lesions, or firm subcutaneous nodules on exposed parts of the body.**

Diagnosis: characteristic **budding cells** seen in KOH preparation of materials from the lesion.

Treatment: IV amphotericin B.

Coccidiomycosis

Caused by *Coccidioides immitis*, a dimorphic fungus acquired by **inhalation of fungal spores.**

The **primary infection is in the lungs**, with **haematogenous spread to the skin, subcutaneous tissue and the bones.**

Skin lesions consist of **panniculitis, abscesses and draining sinuses.**

Treatment: oral ketoconazole or itraconazole.

Mycetoma (Madura foot)

A **chronic infection of the skin, subcutaneous tissue and bones of the feet.**

It takes a **very long time to develop** and is **rare in children.**

The microorganisms are typically **introduced into the body through minor injuries like a splinter or thorn prick**, which is why it is **common among agricultural workers.**

It can be caused by **both fungi and specific types of bacteria:**

- **Eumycetoma** – when the infection is caused by **fungi**
- **Actinomycetoma** – when caused by **filamentous bacteria**

Aetiology:

- **Actinomycetes** such as *Nocardia brasiliensis*, *Actinomadura madurae*, *Streptomyces somaliensis*
- **Fungi** such as *Pseudallescheria boydii*, *Madurella grisea*, or *Madurella mycetomatis*

Treatment: itraconazole.